

SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	July 5	July 20
2021	July 24	July 10
2022	July 13	July 28
2023	July 3	July 17
2024	July 21	July 5
2025	July 10	July 24
2026	July 29	July 14
2027	July 18	July 4

PLANETS

2020: July 14 Jupiter is at opposition, magnitude -2.7 . At midnight, it is visible to the south from northern latitudes and to the north from southern latitudes.

2020: July 20 Saturn is at opposition, magnitude 0.1.

2020: July 22 Mercury is at greatest morning elongation, magnitude 0.5.

2021: July 4 Mercury is at greatest morning elongation, magnitude 0.6.

2021: July 13 Venus and Mars appear one Moon-width apart in the low western dusk sky.

2024: July 22 Mercury is at greatest evening elongation, magnitude 0.7.

2025: July 4 Mercury is at greatest evening elongation, magnitude 0.7.

2027: July 15 Mercury is at greatest morning elongation, magnitude 0.5.

ECLIPSES

2028: July 6–7 A partial eclipse of the Moon is visible across Africa, much of Europe and Asia, Australasia, the Indian Ocean, and Antarctica.

2028: July 22 A total eclipse of the Sun is visible across Australasia and the Indian Ocean, with a partial eclipse visible in southeast Asia.

JULY

The strong man of Greek mythology, Hercules, lies overhead as seen from mid-northern latitudes, between the bright stars Vega (in Lyra) and Arcturus (in Boötes). South of Hercules is another large constellation, Ophiuchus, which represents a man encircled by a serpent, Serpens. In southern skies, the Milky Way passes overhead from the southwest to the northeast. The zodiacal constellations Scorpius and Sagittarius stand high in the Milky Way's richest part.

NORTHERN LATITUDES

THE STARS

Overhead lies Hercules, which is a large but not particularly striking constellation. Its most distinctive feature is a quadrangle formed by four stars, called the Keystone. North of Hercules lies the lozenge-shaped head of Draco, the Dragon. Between Draco and the north celestial pole is the bowl of the Little Dipper, in Ursa Minor.

Arcturus, in Boötes, remains prominent in the western sky. Spica, in Virgo, is lower in the southwest, and the Big Dipper dips low in the northwest. In the eastern half

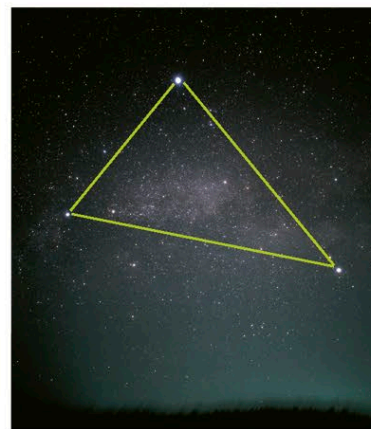
of the sky, the stars of the Summer Triangle climb ever higher, while the Square of Pegasus appears closer to the eastern horizon.

Low in the south are the rich constellations Scorpius and Sagittarius. This is the best month for northern observers to see the two most southerly zodiacal figures in the evening sky.

DEEP-SKY OBJECTS

Ophiuchus, the large constellation between Hercules and Scorpius, contains numerous globular clusters, although only two of them, M10 and M12, are of any note. The most impressive deep-sky objects in Ophiuchus are the open clusters IC 4665 and NGC 6633, both

good binocular sights. The globular clusters M13, in Hercules, and M5, in the head of Serpens, remain well positioned this month.



THE SUMMER TRIANGLE
Deneb (left), Vega (top), and Altair (right) form a prominent triangle that remains visible well into fall in northern skies.

9 A.M.

6 A.M.

3 A.M.

MIDNIGHT

